Journal of Asian Earth Sciences 201 (2020) 104517

Contents lists available at ScienceDirect

Journal of Asian Earth Sciences

journal homepage: www.elsevier.com/locate/jseaes

Paleomagnetism and rock magnetism of early Cretaceous Rajmahal basalts, NE India: Implications for paleogeography of the Indian subcontinent and migration of the Kerguelen hotspot

M.R. Kapawar, Venkateshwarlu Mamilla*

CSIR- National Geophysical Research Institute, Hyderabad-500007, India

ARTICLE INFO

Keywords:
Paleomagnetism
Rock magnetism
Rajmahal Traps
Kerguelen hotspot
APWP
Paleogeographic position

ABSTRACT

We present here new paleomagnetic and rock magnetic results from Rajmahal Traps (RT), India. The results are comprising of 350 standard specimens collected from 18 sites (90 samples). The detailed rock magnetic studies indicate that a Single Domain (SD) magnetite [Fe_3O_4] is the dominant magnetic carrier with ilmenite, pyrrhotite, and maghemite in the accessory form. The calculated Characteristic Remanent Magnetization (ChRM) directions yield a mean declination of 317.51°; mean inclination of −61.92° (α_{95} = 3.6°; k = 93.39). The obtained Virtual Geomagnetic Pole (VGP) position of the RT at ~117 Ma is 11.37°N, 297.58°E (62.42°W) with A95 = 3.5°, k = 99.99, dp = 4.33, dm = 5.58, and the calculated paleolatitude is at 43.1°S indistinguishable from that of earlier works. This paleomagnetic pole of RT basalts has minimal effect of secular variation and comply with the Geocentric Axial Dipole (GAD) model. The VGP is plotted on synthetic Apparent Polar Wander Path (APWP) for India, constrained better, and agrees well with the previously reported poles. The postulation of migration of mantle plume and the global mantle circulations are tested, and the present study results are in accord with this, suggesting the southward migration of the Kerguelen hotspot (mantle plume) by at least 6°.

1. Introduction

The Cretaceous plate reconstruction studies (e.g., Davies et al., 1989; Royer and Coffin, 1992) suggest that Rajmahal lavas form a part of the Large Igneous Province (LIP), which includes the Naturaliste-Bunbury basalts of Australia and southern and middle Kerguelen Plateau lavas. The Rajmahal Traps (RT) is situated on the eastern margin of India and a constituent of the Rajmahal- Bengal- Sylhet (RBS) volcanic outpouring on the Indian subcontinent. At ~ 130 Ma, the Indian Plate broke off from Australian Plate (Mihut and Muller, 1998; Heine et al., 2004; Torsvik et al., 2008), and drifted northwards (Bian et al., 2019). The initial Kerguelen hotspot (mantle plume) activity has triggered this break-up (Mahoney et al., 1983; Kent, 1991; Frey et al., 2000).

The paleomagnetic studies on RT basalts were initiated by Clegg et al. (1958), Radhakrishnamurthy (1963, 1970), and McDougall and McElhinny (1970). Some workers have reported the magnetic polarity reversal in RT (Klootwijk, 1971; Ghosh, 1979; Bagchi, 1991; Sherwood et al., 1993). Subsequently, the paleomagnetic approach for RT basalts was followed by Rao et al. (1996); Rao and Rao (1996). A paleomagnetic study over the Marion and the Reunion mantle plume related magmatic rocks of India (Radhakrishna and Joseph, 2012) suggested southward and northward migration of these plumes with geologic time. However, other results from Marion hotspot products in Madagascar report the stationary Marion hotspot over time (Meert and Tamrat, 2006). Further, the previous rock magnetic studies from the RT basalts (Radhakrishnamurthy et al., 1977, 1991; Sherwood et al., 1993; Sherwood and Mallik, 1996) suggested that titanomagnetites and ilmenites dominate the main magnetic mineralogy. We have conducted the paleomagnetic and rock magnetic studies on RT basalts in order to contribute new results towards the global paleomagnetic dataset and to assess the polarity reversal as reported by earlier studies. We have plotted a refined pole position on Indian APWP, and tested whether the Kerguelen hotspot (mantle plume) migrated over geologic time.

2. Geological setting and sampling

The RT basalts are exposed from 24°05′N to 25°15′N and 87°24′E to 87°51′E (Fig. 1). They extend over a north–south elongated area of about 4500 km^2, which falls mainly to the south of the Ganga river plains in eastern India. The present study focuses on RT basalts in the regions of Sahibganj and Pakur districts of Jharkhand state, India. The youngest formation observed here is the alluvium cover and laterite

* Corresponding author.
E-mail address: mvwarlu@ngri.res.in (V. Mamilla).

https://doi.org/10.1016/j.jseaes.2020.104517
Received 19 May 2020; Received in revised form 4 August 2020; Accepted 12 August 2020
Available online 17 August 2020
1367-9120/ © 2020 Elsevier Ltd. All rights reserved.

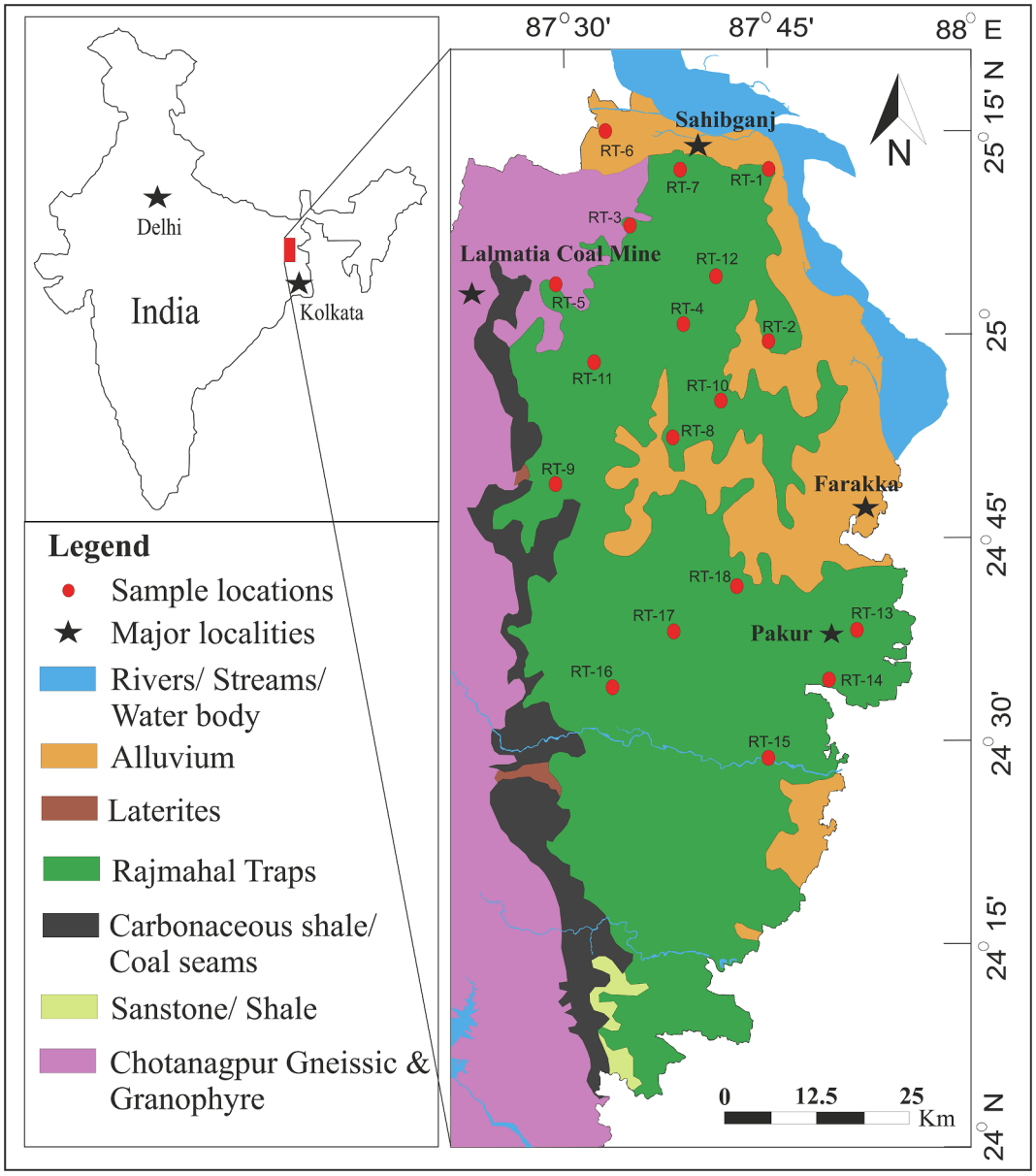


Fig. 1. Schematic geological map of the Rajmahal Traps (RT) basalts. Red circles are the sampling locations near Sahibganj and Pakur districts (modified after Dept. of Mines and Geology, Geological Survey of India). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

capping due to the fluvial action of river Ganga and Brahmaputra. The very first attempt to map the geology of RT was made by Ball (1877) and later by Roy Chowdhury (1965). The RT basalts are tholeiitic lava sequences having a thickness of around 600 m (McDougall and McElhinny, 1970; Klootwijk, 1971) and are separated by intertrappean sandstones and shales (Tiwari and Jassal, 2003). In the stratigraphic sequence, the RT basalts overlie the older Dubrajpur Formation comprising sandstones and shales. At least ten distinct lava flows have been identified by Hobson (1929), and individual flows range from 20 to 75 m in thickness, while Pascoe (1959) speculated that the number of flows could be more than earlier believed. The RT basalts are mostly compact, but some samples have minute amygdales of calcite, analsite, chalcedony, and agate (Ball, 1877).

The geochemical and geochronological studies of RT (Baksi et al., 1987; Ghatak and Basu, 2013) suggest their origin can be attributed to the Kerguelen hotspot. However, Mahoney et al. (1983) reported that the Kerguelen hotspot may not be directly involved but provided the heat for Rajmahal activity. It was later suggested that along with the elevated Indian Ocean lithosphere, the Rajmahal flood basalts are one of the Kerguelen hotspot related magmatic provinces left on the eastern margin of the Indian Plate (Luyendyk and Rennick, 1977; Morgan, 1981; Duncan, 1981). The plate tectonic histories of volcanic outputs like the Ninetyeast Ridge, the Kerguelen Plateau, Broken Ridge, and the Rajmahal flood basalts suggest they belong to the Kerguelen hotspot at appropriate times during last 120 million years (Davies et al., 1989).

The sampling locations are represented in Fig. 1. Around 90 oriented basalt blocks from 18 sites were collected from horizontal strata, using the Brunton compass. The columnar jointing was exposed at a few of the quarry sections with 15–20 cm polygons. The physical observation (thickness in particular) of the flows was not possible at every location of sampling, but as evident from one locality, it was ~ 20 m thick and at the base, sandstone was exposed. A minimum of 5 fresh and unaltered oriented samples were collected from each site. The samples collected were later drilled and cut to get 350 standard specimens.

3. Methods

The Natural Remanent Magnetization (NRM) was measured on all discrete specimens, using AGICO's JR-6 spinner magnetometer (Czech Republic) with a signal to noise ratio of 2.4 μA/m. The bulk magnetic susceptibilities (k) were measured using AGICO's MFK1-FA (Czech Republic) susceptibility apparatus, at frequency 976 Hz. The stability of the NRM directions of samples was tested by the application of stepwise Alternating Field Demagnetization (AFD) and Thermal Demagnetization (ThD), using Molspin AF- demagnetizer (Magnetic Measurements, U.K.) and MMTD- 80 thermal demagnetizer (Magnetic Measurements, U.K.), respectively.

Rock magnetic studies, such as thermoremanent curves, hysteresis loops, Isothermal Remanent Magnetization (IRM) acquisition curves, and backfield coercivity spectra were conducted to identify the magnetic mineralogy, grain size, domain state, and their thermomagnetic stability, using an Advanced Variable Field Translation Balance (AVFTB). The powdered samples weighing around 400–600 mg were used for the rock magnetic study.

All the paleomagnetic and rock magnetic measurements were carried out at the Paleomagnetism Laboratory of CSIR- National Geophysical Research Institute (NGRI), Hyderabad, India.

4. Results

4.1. Paleomagnetism

The NRM intensity values of RT basalts range between 0.231 and 0.028 A/m, and bulk magnetic susceptibility values range between 1.001×10^{-2} to 9.978×10^{-3}SI units with an average of 4.412×10^{-3}SI units. The paleomagnetic investigations comprising AF and thermal demagnetizations yielded a very reliable and stable dataset. AF demagnetization on 172 standard specimens was carried out in 15 successive steps and, thermal demagnetization was performed on 173 standard specimens in 11 progressive demagnetization steps. After each demagnetization step, the remanence was measured using the JR-6 spinner magnetometer. Fig. 2 shows the representative Zijderveld diagrams and intensity decay curves for both AF (Fig. 2a, 2b) and thermal (Fig. 2c, 2d) specimens. A weak viscous magnetization component was erased at around 15–30 mT of the applied magnetic field and ~ 200 °C temperature, but is not the case for all the specimens measured. The High Field Component (HFC) and High Temperature Component (HTC) ascertained, which decay towards the origin after ~ 30 mT and ~ 300 °C, respectively, of total demagnetization. For both HFC and HTC determinations, 6 to 8 sequential demagnetization steps that decayed steadily towards the origin, were used to define Characteristic Remanent Magnetization (ChRM) directions. The Zijderveld vector migration diagrams (Zijderveld, 1967) were constructed to display and assess the demagnetization trends for each specimen. Since all the samples are from horizontal strata, it is presumed that tilt correction is not required.

The ChRM directions of all the specimens were calculated by Principal Component Analysis (PCA) of a best-fit line (Kirschvink, 1980) using Remasoft-3.0 software, and the mean direction was determined using Fisherian statistics (Fisher, 1953). The ChRM direction is considered to be the most stable magnetization component, which eventually yield better results about paleomagnetic stability. The paleomagnetic dataset, derived from this study yielded good grouping, as shown in the stereonet plot (Fig. 3), and all the sites have stable and consistent paleomagnetic directions. For the ChRM calculations, each discrete specimen's data acceptance limits were constrained within MAD ≤ 5° and $\alpha_{95} \leq 5°$.

The mean ChRM direction is D = 317.51°; I = -61.92° (α_{95} = 3.6°; k = 93.39). The obtained mean upward dip (-61.92°) is in accord with the postulation that the Indian subcontinent/ plate was situated in the southern equatorial position (43.1°S) at the time of formation of these volcanic rocks. The corresponding Virtual Geomagnetic Pole (VGP) is at 11.37°N and 297.58°E (A95 = 3.5°, k = 99.99, dp = 4.33, dm = 5.58), considering mean site latitude 25° 10′ and longitude 87° 40′. The 0 to 120 Ma synthetic Apparent Polar Wander Path (APWP) of India, deduced from the Global Apparent Polar Wander Path (GAPWaP; Torsvik et al., 2012), was considered for plotting the VGP of RT basalts. The Deccan Super Pole (DSP; Vandamme et al., 1991), the VGP pole from Rajmahal Traps (McDougall and McElhinny, 1970; Klootwijk, 1971) and Sylhet Traps (Athavale et al., 1963; Rao et al., 1993) are also plotted, as reference. The detailed paleomagnetic results are provided in Table 1. The paleomagnetic results yield good grouping of directions from all the studied sites, suggesting short time lava/ magma emplacement.

4.2. Rock magnetism

The rock magnetism is a non-destructive and easy-access method, useful in identifying the magnetic mineralogy. Rock magnetic analyses (thermoremanent curves, hysteresis loops, IRM acquisition curves, and coercivity spectra) were performed to understand the magnetic mineralogy, curie temperatures (Tc), mineral grain sizes, and domain state of samples. Our study presents more detailed rock magnetic results that expand on preliminary results of Sherwood (1992). Sherwood and Mallik, (1996) discussed the discrepancies in their results caused by electronic instability of classical instruments during measurements. An attempt is made in the present study to generate a more refined dataset using the available modern and sophisticated instrument, i.e., AVFTB.

4.2.1. Thermoremanent curves

The thermoremanent curves can be used to identify different types of magnetic materials (Dunlop and Ozdemir, 1997). Fig. 4 shows the thermoremanent curves for RT basalts. None of our samples show the reversible nature of thermoremanent curves, which suggests transformation of magnetic minerals from unstable to stable magnetic phases. A sharp magnetic susceptibility increase in the cooling curves between 570 and 600 °C is observed in almost all the samples (Fig. 4), suggesting that the bulk magnetic mineralogy dominated by ferrimagnetic minerals like magnetite (Dunlop and Ozdemir, 1997). Nearly all the samples have experienced low-temperature oxidation (maghematization) during the experiment.

In RT 1.2 and RT 7.1 (Fig. 4), the cooling curves show higher susceptibility than the heating curves, with a sharp increase at ~ 580 °C, indicating magnetite formation. Further, samples RT 1.2, RT 3.4, and RT 7.1 yield a substantial susceptibility drop in the heating cycle between ~ 200 °C-300 °C, indicating a temperature contemporaneous with the mineral ilmenite and is eventually transformed into a stable magnetic phase, i.e., magnetite.

The heating cycle of RT 2.2, RT 5.5, and to a smaller extent RT 4.2 (Fig. 4), reflects an existence of monoclinic pyrrhotite (Tc = ~320 °C), which is completely transformed to magnetite (Dunlop and Ozdemir, 1997) upon heating. All these samples show zero susceptibility at ~ 650 °C during heating cycle, this is not observed in cooling cycle, indicates the presence of low stability phase of maghemite. The maghemite (Tc = 590–675 °C) seems to appear in all the samples either as a product of maghematization, cooling rate variations, or mild hydrothermal alterations (Dunlop and Ozdemir, 1997). The samples RT 13.4, RT 15.4, and RT 17.3 suggest magnetite's presence in both heating and cooling curves with a little amount of maghematization at the end of the heating cycle. The heating cycles of these samples also show the dissolution of magnetic minerals from the Titanomagnetite Solid Solution (TMS) series (as appeared between a short interval of 300–400 °C) to magnetite, with heating. The oxidation in RT basalts could be due to hydrothermal alteration in the past or a recent weathering mechanism (Sherwood, 1992).

The thermoremanent experiment indicates two principal iron oxide phases, i.e., low Ti magnetite with Tc = 525–570 °C and maghemite

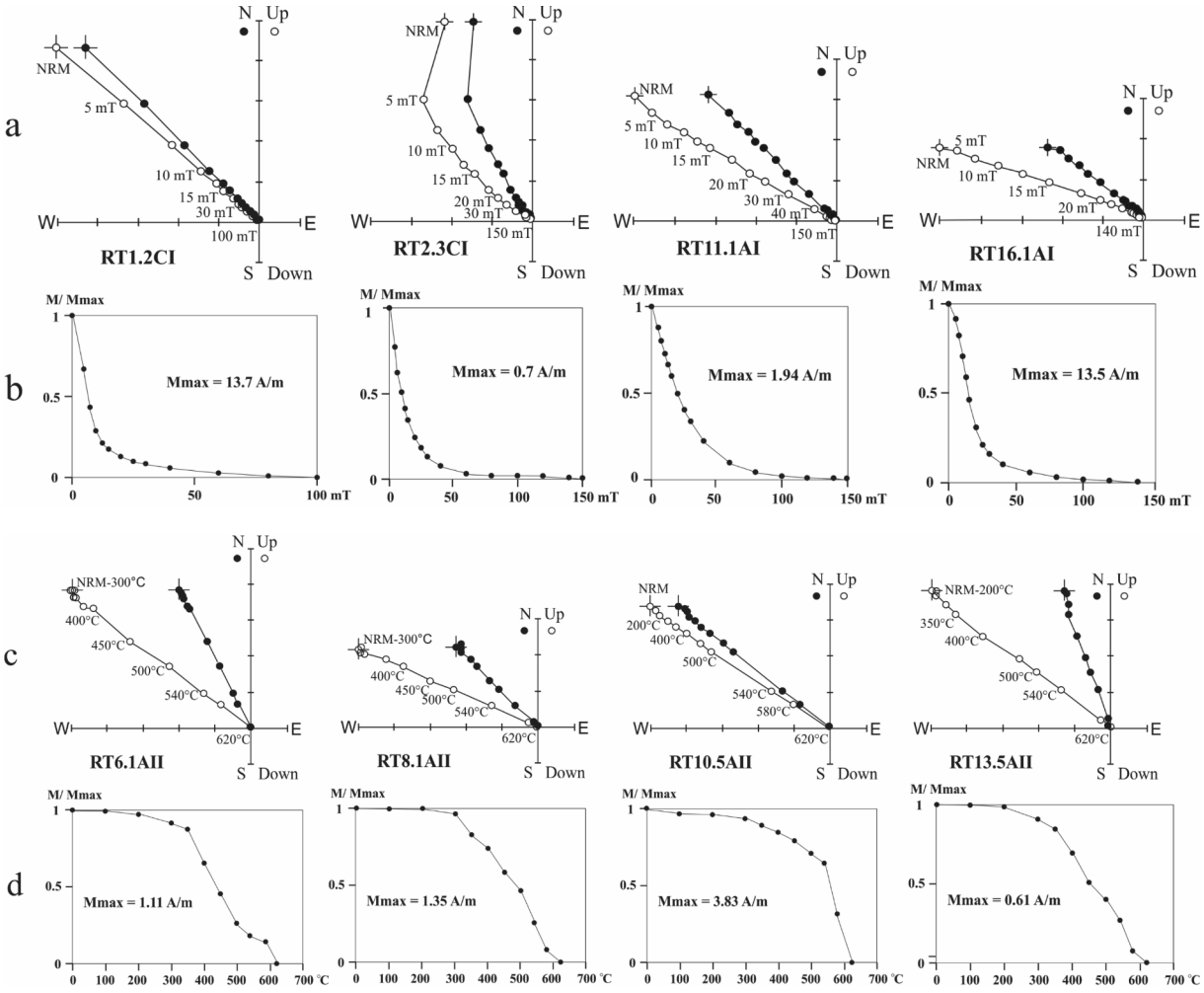


Fig. 2. The orthogonal projections and intensity decay curves plot for AF (a, b) and Thermal (c, d) demagnetizations. Solid (open) data points representing vector endpoints projected onto the horizontal (vertical) plane.

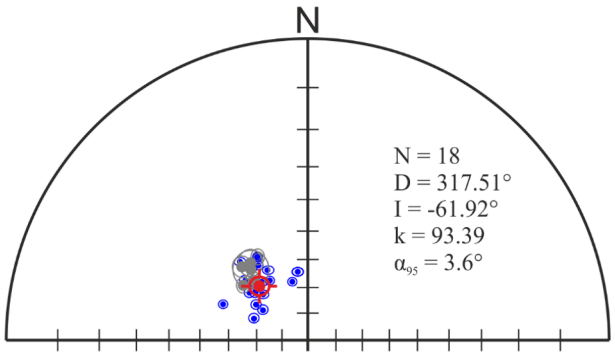


Fig. 3. The stereoplot is showing the paleomagnetic data. The site mean directions (blue) with 95% confidence circle on mean remanence (α_{95}) and the published results in grey. The grand mean direction with α_{95} is shown in red. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

with Tc = 610–640 °C. Moreover, the heating curves with Tc below 350 °C could indicate that the magnetic susceptibility is prevailed by ilmenite and ferrimagnetic pyrrhotite, which is later controlled by ferrimagnetic grains, especially magnetite. Two stages of oxidation appear in the thermoremanent curves, one during heating and the other in cooling. The former suggests the maghematization of magnetic minerals, and the latter is the sub-solidus oxidation of minerals.

4.2.2. Hysteresis loop study

The magnetic hysteresis loop analysis was conducted on 18 samples. The magnetic hysteresis loops obtained for these samples are presented in Fig. 5. The samples with different mineral assemblages and dissimilar grain sizes showcase different shapes of magnetic hysteresis loops. Further, the remanence ratio (Mrs/Ms) and coercivity ratio (Hcr/Hc) were determined from the hysteresis parameters of saturation magnetization (Ms), saturation remanence (Mrs), coercivity force (Hc) and coercivity remanence (Hcr), and summarised in table 2. These ratios (Mrs/Ms versus Hcr/Hc) are plotted on a 'Day plot' (Day et al., 1977), shown in Fig. 6. This Day plot specifies and distinguishes the Single Domain (SD), Pseudo-Single Domain (PSD), and Multidomain (MD) magnetic grains and will also serve as a better magnetic granulometry indicator when plotted together with theoretical mixing curves (Dunlop, 2002).

The samples have Mrs/ Ms ranging from 0.036 to 0.309, with an average of 0.169 and Hcr/ Hc ranging from 1.06 to 1.43 with an average of 1.23. Fig. 5 shows that all the loops are fully saturated by a maximum of 500 mT in an applied field of 1 T, showing S-curve loops. The samples RT 2.2, RT 3.4, RT 5.5, and RT 16.3 show similar but wider hysteresis loops with a saturation near 500 mT and exhibit the presence of a substantially-high coercivity component but not dominant. The samples RT 9.2, RT 10.4, RT 11.5, RT 13.4, and RT 18.4 display consistently look-alike hysteresis loops in the near wasp-waisted shape due to medium to fine-grain size magnetic minerals. These results reveal that most of the recorded remanence is held by grains falling within the

Table 1
Concise paleomagnetic data from the Rajmahal Traps (RT) basalts.

Site	Rock type	Lat (°N)	Long (°E)	N/n	D (°)	I (°)	α_{95}	k	MAD	VGP Lat (°N)	VGP Long (°E)	dp	dm
RT-1	Basalt	25° 13′	87° 45′	19/14	292	−55	1.7	31.7	1.7	2.00	317.30	11.79	16.76
RT-2	Basalt	24° 59′	87° 44′	18/15	327	−56	1.5	39.4	1.2	18.12	298.12	11.01	15.14
RT-3	Basalt	25° 1′	87° 33′	20/16	320	−62	2.2	21.1	2.1	13.61	295.94	22.39	29.10
RT-4	Basalt	25° 20′	87° 37′	20/17	317	−61	1.2	57.1	0.9	14.63	300.98	12.35	16.76
RT-5	Basalt	25° 5′	87° 29′	20/14	346	−67	1.2	56.0	1.6	14.77	276.19	14.73	17.84
RT-6	Basalt	25° 15′	87° 32′	19/15	329	−60	1.1	72.5	1.2	17.60	291.80	11.67	15.37
RT-7	Basalt	25° 13′	87° 36′	16/13	327	−64	1.5	51.8	1.7	13.68	291.99	17.40	22.10
RT-8	Basalt	24° 49′	87° 35′	19/13	304	−66	1.1	76.6	1.2	3.20	300.63	12.51	15.46
RT-9	Basalt	24° 48′	87° 29′	20/13	292	−68	0.8	142.2	2.7	−6.58	305.04	9.15	11.04
RT-10	Basalt	24° 53′	87° 44′	20/17	318	−52	1.2	61.9	1.3	21.21	304.34	9.88	14.63
RT-11	Basalt	24° 56′	87° 34′	20/14	341	−64	2.1	18.9	1.5	19.97	282.74	28.49	36.95
RT-12	Basalt	25° 3′	87° 42′	20/16	312	−64	0.9	88.4	2.2	8.74	299.37	11.87	15.15
RT-13	Basalt	24° 36′	87° 51′	20/15	328	−55	1.6	38.6	1.2	24.22	290.26	9.36	13.53
RT-14	Basalt	24° 33′	87° 49′	16/12	309	−62	1.2	90.1	1.2	6.12	303.20	11.38	14.63
RT-15	Basalt	24° 28′	87° 45′	20/15	302	−69	1.7	32.4	1.1	2.01	304.82	10.58	13.40
RT-16	Basalt	24° 32′	87° 32′	19/17	313	−58	1.2	80.8	1.5	10.47	302.64	8.86	11.74
RT-17	Basalt	24° 40′	87° 36′	19/15	323	−56	2.7	19.4	1.1	22.37	301.23	9.03	14.13
RT-18	Basalt	24° 42′	87° 44′	20/15	311	−63	1.3	50.2	0.8	9.31	301.66	11.73	15.29
Grand Mean	**345 / 266**	**317.51**	**−61.92**	**3.6**	**93.39**	**1.5**	**11.37**	**297.58**	**4.33**	**5.58**			

Note: Lat (°N) and Long (°E) are site latitude and site longitude respectively; N and n is number of specimens analyzed from a site and number of specimens considered for final calculations; D(°) and I(°) are the mean remanence declination and mean remanence inclination; α_{95} = 95% confidence circle on mean remanence; k is the Fisher (1953) precision parameter; MAD is the maximum angular deviation; VGP Lat and VGP Long are the VGP latitudes and VGP longitudes; dp and dm are the Semi-axes of the oval of 95% confidence about the VGP.

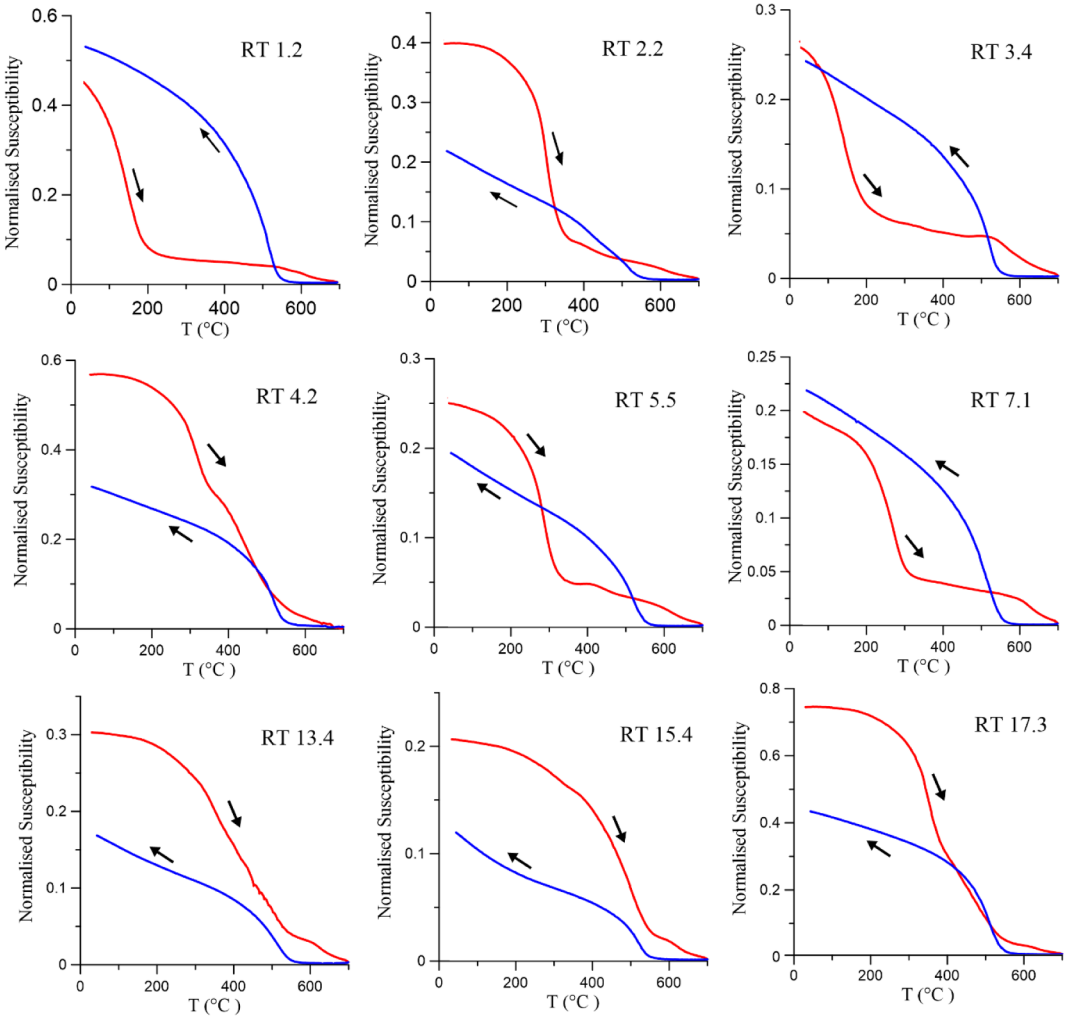


Fig. 4. Thermoremanent curves of selected samples that were heated from room temperature to 700 °C and cooled back to room temperature in the presence of 100 mT of constant field. The red (blue) lines indicate the heating (cooling) cycle of analysis. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

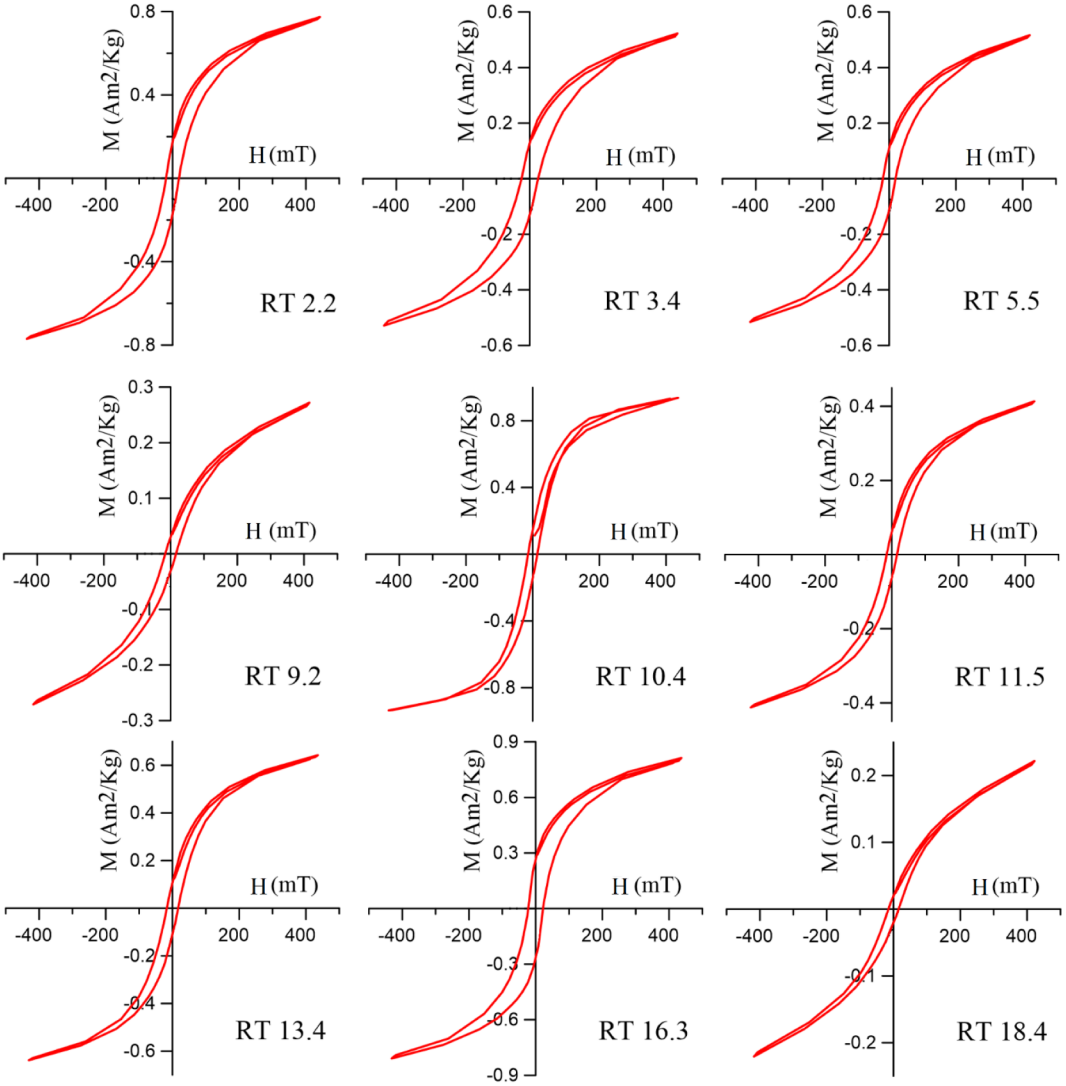


Fig. 5. Magnetic hysteresis loops of selected RT basalt samples.

Table 2
The hysteresis parameters derived from hysteresis loops of samples.

Sample ID	Mrs (Am^2/kg)	Ms (Am^2/kg)	Mrs/Ms	Hcr (mT)	Hc (mT)	Hcr/Hc
RT 1.1	0.29	0.98	0.295	23	20	1.15
RT 2.2	0.19	0.94	0.202	22	20	1.1
RT 3.4	0.15	0.66	0.227	30	25	1.2
RT 4.2	0.20	1.14	0.175	18	15	1.2
RT 5.5	0.13	0.68	0.191	20	14	1.43
RT 6.2	0.05	0.450	0.111	18	12	1.5
RT 7.1	0.15	0.55	0.100	33	31	1.06
RT 8.2	0.05	0.45	0.111	19	17	1.11
RT 9.2	0.04	0.40	0.100	19	15	1.26
RT 10.4	0.20	1.07	0.186	17	12	1.41
RT 11.5	0.075	0.55	0.136	18	15	1.2
RT 12.4	0.17	0.80	0.212	17	15	1.13
RT 13.4	0.13	0.80	0.162	19	15	1.26
RT 14.5	0.10	0.70	0.142	18	15	1.2
RT 15.4	0.08	0.63	0.126	15	11	1.36
RT 16.3	0.3	0.97	0.309	22	20	1.1
RT 17.3	0.35	1.53	0.228	19	15	1.26
RT 18.4	0.013	0.36	0.036	19	14	1.35

SD region (Fig. 6), as is typical for natural volcanic rock samples.

4.2.3. IRM acquisition curves and coercivity spectra

The IRM acquisition curves are significant in identifying the saturation magnetisation of different rock types. The maximum applied magnetic field is 1 Tesla. All the samples show a significant rise in IRM acquisition curves with a saturation at ~ 300 mT, except few at ~ 400 mT. The coercivity spectra for the same set of samples show coercivity values ranging between 30 and 62 mT, which deciphers the predominance of low coercive magnetic mineral in the samples. Both IRM acquisition curves and coercivity spectra (Fig. 7) suggest that the resultant remanent magnetization is dominated by low coercivity magnetic mineral, and which is from the TMS series. The typical saturation of IRM curves at ~ 300 mT, coercivity < 70 mT, and the SD grains, which have generated the symmetrical curves, due to the presence of magnetite. The present study findings differ a little from the previous research, suggesting the main magnetic minerals in the samples are Ti-rich titanomagnetites (Sherwood, 1992; Sherwood and Mallik, 1996) with varying domain states.

Overall, the main magnetic carrier identified in our study is the ferrimagnetic magnetite and other minerals, like ilmenite, pyrrhotite, and maghemite, revealing their accessory occurrence in the samples. Identifying these magnetic minerals in RT basalts may allow us to use mineral alteration as a reference for further studies concentrating on

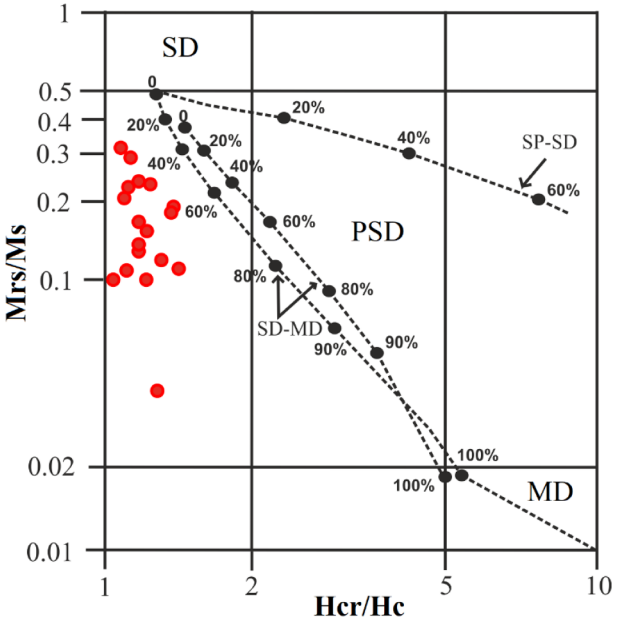


Fig. 6. The hysteresis parameters, Mrs/ Ms and Hcr/ Hc are plotted on a Day plot (Day et al., 1977) with the theoretical SD-MD and SP-SD mixing lines from Dunlop (2002).

the geological processes. The rock magnetic properties of the RT basalts suggest that the samples have minor variations, which is exclusively controlled by the cooling history of lava.

5. Discussion

5.1. Stability of ChRM directions and comparison with earlier studies

Both demagnetization techniques (AF and thermal) were executed on 350 standard specimens to get the precise ChRM directions. A strict Fisher statistics (MAD $\leq 5^{\circ}$ and $\alpha_{95} \leq 5^{\circ}$) was adopted for ChRM calculations, which has yielded quality results from these outcrops. The ChRM directions reported in the present study are mostly attained at 40–100 mT of peak AF fields and 400–580 °C thermal steps. Our study shows a Van der Voo (1990) data quality criteria (Q-value) of 6, which is quite good for any paleomagnetic study. These results are similar to Rao and Rao (1996), Sherwood and Mallik (1996), and Bian et al. (2019).

The paleomagnetic results of ODP site 1136 (119 Ma), 1137 (108 Ma), 1138 (100 Ma), 1141 (95 Ma), and 1142 (95 Ma) depict normal polarity, which is consistent with our new results indicating the acquisition of magnetization during the Cretaceous Normal Superchron (CNS; 121–83 Ma). Coffin et al. (2002), reported the higher magma eruption rate from the Kerguelen hotspot that caused rapid emplacement of RT basalts. Since there is not much inclination difference among the sites (and among previous studies; Table 3), this eventually minimizes the effect of secular variation on these rocks. If any, the secular variations are averaged out by taking a 'sufficient number' of samples (Deenen et al., 2011), and such a paleopole calculated complies with the Geocentric Axial Dipole (GAD) model. This feature emphasizes the relatively fast consolidation of basaltic rocks after the eruption from the Kerguelen hotspot (mantle plume), which confirms the ^{40}Ar- ^{39}Ar results of Baksi (1995). It is significant to note that the magnetic polarity reversal reported earlier (Klootwijk, 1971; Rao et al., 1993, 1996; Das et al., 1996; Sherwood and Mallik, 1996) has not been observed in the present study. Two sites RJ2 and RJ10 at Gandeswari Hill of Sherwood and Mallik (1996), reported reverse- intermediate directions as D = 93.3°, I = -57.6°, and D = 234°, I = 40.7° respectively, and suggested the need of more samples to confirm the reverse polarity. Whereas in the present study, we have collected samples from the same area (RT 5), which show a clear normal polarity direction (D = 346°, I = -67°), so there ends the doubt of the presence of reversed polarity in this area. However, Das et al. (1996) have reported a brief reversal event in the Hetbanda section of RT basalts, which he claims as to the commencement of trap emplacement. From our results, it is established that the basalts with normal polarity exist in the northern parts of RT.

5.2. Paleogeography of the Indian subcontinent

The paleogeography of the Indian subcontinent has been attempted by assessing the volcanic outputs left on the Indian Plate by the Indian Ocean hotspots. The Indian Ocean has mainly four hotspots (plume heads), the Kerguelen, Reunion, Marion, and Crozet. The present study concerns with a volcanic product of the Kerguelen hotspot, i.e., Rajmahal basalts. The Rajmahal basalts of the Indian subcontinent and the Bunbury basalts of the Australian Plate (Olierook et al., 2016) has preserved the geological connectivity between these two continental plates and suggest their genesis through a single parental hotspot, i.e., the Kerguelen hotspot. It indicates that Indian and Australian plates were nearby when these rock units were emplaced on respective plates due to initial Kerguelen volcanic activity. At ~ 117 Ma, the RT was formed over the eastern margin of India due to Kerguelen hotspot activity (Courtillot, 1999; Coffin et al., 2002; Kent et al., 2002; Ghatak and Basu, 2013). Most of the northern Indian rocks are either eroded/ buried under extensive alluvium cover or are disturbed due to orogeny and the Himalayan subduction mechanism. Hence, the RT becomes a suitable candidate for study to decipher the Indian subcontinent paleogeographic position during the early Cretaceous period.

In this paper, we present a new paleomagnetic pole from RT to better define the paleogeographic position of the Indian Plate (Fig. 8a), in the early Cretaceous period. Moreover, for comparison the Indian, Australian, and Antarctica plates are brought closer but within acceptable limits as per the present paleomagnetic results and geologic evidence. This is a common practice to show different continental blocks in a single meaningful configuration to define paleogeography. Between 117 and 130 Ma time window, the Indian, Australian, and Antarctica plates were started to separate, with India Plate speedily drifted northwards. Around 99.6 Ma, India- Seychelles- Madagascar was completely separated from Australia- Antarctica and brought the Kerguelen Plateau to be exposed subaerially (Coffin, 1992; Frey et al., 2000). Subsequently, the Indian subcontinent stayed as islandic mass for about 30 Ma before its passage over the Reunion hotspot at ~ 65 Ma, which led to the emplacement of the Deccan basalts on the western margin of India. The Indian subcontinent had both southern and northern connections with the Chagos-Laccadive Ridge and the Burma- Sumatra, respectively, at around ~ 55 Ma (Aitchison et al., 2007), which has been considered as the actual collision with the Eurasian Plate (southern Tibet). After its break-up from eastern Gondwana, the Indian Plate had attained a very high speed of 18–20 cm/yr during the early Cretaceous period and then decreased to ~ 5 cm/ yr at ~ 50 Ma as it amalgamated with the Eurasian Plate (Klootwijk et al., 1992; Gaina et al., 2007), and then the oceanic Greater India basin closed at ~ 25 Ma, which marked the completion of the India-Asia collision (Qin et al., 2019).

5.3. APWP for India and migration of the Kerguelen hotspot

The VGP for the RT basalts yield a paleopole at 11.37°N, 297.58°E, which is not significantly different from the previous studies (Table 3). However, our data present the best reliability for further interpretations alike Sherwood and Basu Mallik (1996), in constructing the APWP for India and discussing the Kerguelen hotspot's migration. The VGP is plotted on the 0–120 Ma synthetic APWP for India derived from the GAPWaP by Torsvik et al. (2012) (Fig. 8b), which shows perfect fall on the synthetic APWP.

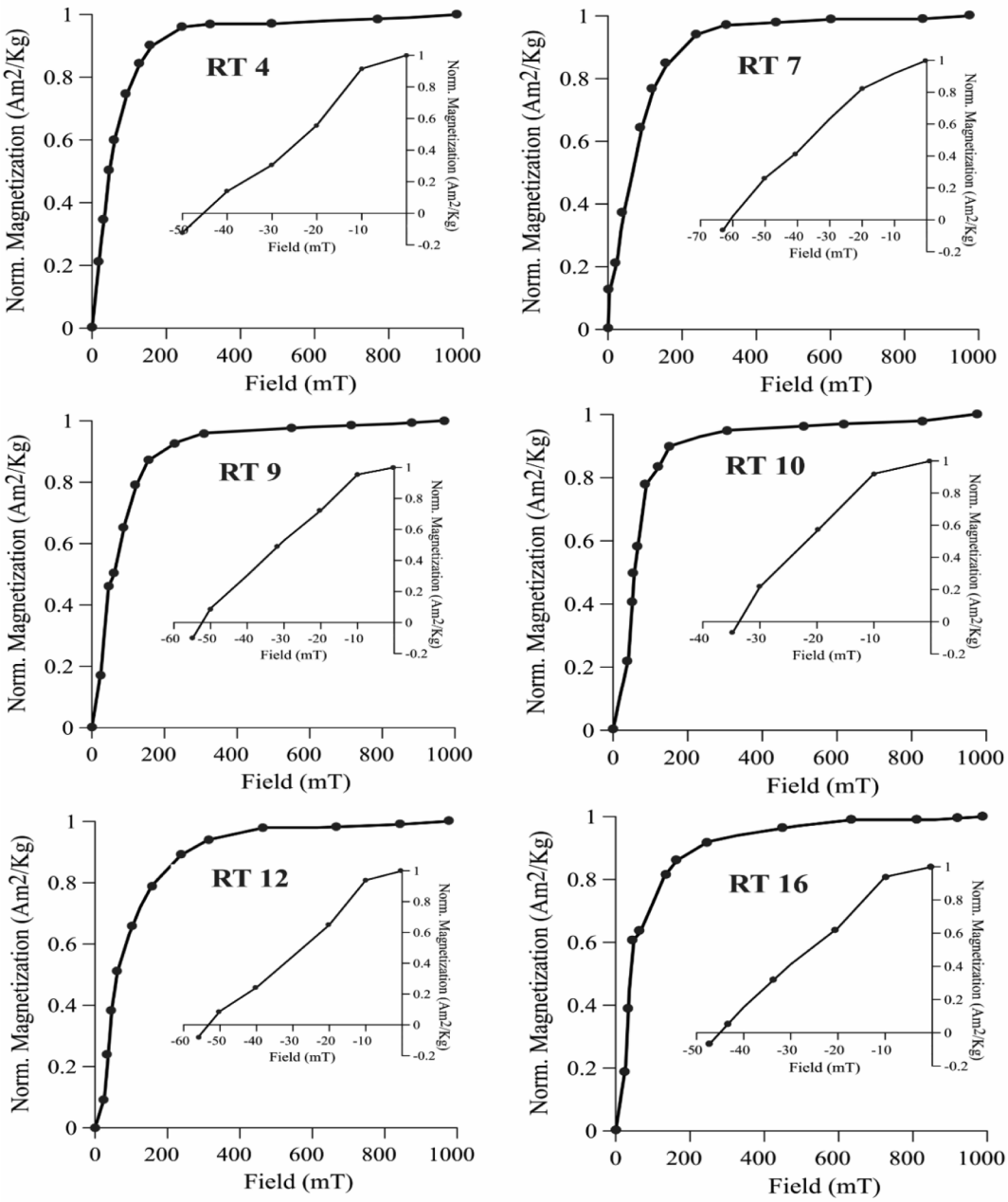


Fig. 7. Isothermal remanent magnetization (IRM) acquisition curves and coercivity spectra for selected samples.

Table 3
The comparison of mean directions obtained from previous studies of Rajmahal and Sylhet Traps (of RBS volcanic province) with the data of the present study.

Sr. No.	Formation	Site Coordinates		D (°)	I (°)	α_{95} (°)	VGP		References
		Lat (N)	Long (E)				Lat (N)	Long (W)	
1	Rajmahal Traps	25°	87° 51′	328	−64	2 (α50)	13°	70°	Clegg et al., 1958
2		–	–	320.6	−65.2	6	10°	67°	Rao et al., 1996
3		–	–	315	−65	4	7°	63°	Klootwijk, 1971
4		–	–	323	−64	4	12°	67°	Radhakrishnamurty, 1970
5		–	–	310	−67	4	3°	62°	McDougall and McElhinny, 1970
		–	–	317.8	−63.2	3.4	10.4°	63.4°	Sherwood and Basu Mallik, 1996
6	Sylhet Traps	25°	91°	332	−59	7	16°	60°	Athavale et al., 1963
7		–	–	319 & 235	−62 & −62	7 & 5	12° & -	59° & -	Rao et al., 1993
8		25° 13′	91° 43′	255	−57	3.44	26.79°	29.25°	Kapawar and Venkateshwarlu, 2019
9	Rajmahal Traps	25° 10′	87° 40′	317.51	−61.92	3.6	11.37°	62.42° (297.58°E)	Present study

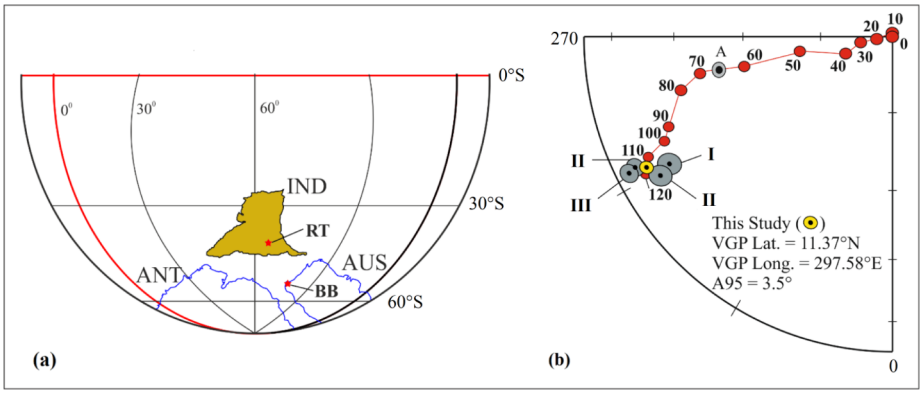


Fig. 8. (a) The paleogeographic position of the Indian subcontinent at ~ 117 Ma (Early Cretaceous) determined from the present study. The RT denotes Rajmahal Traps near the eastern margin of the Indian subcontinent, BB corresponds to the Bunbury basalts at the western margin of Australia, and a paleolatitude of 52 ± 4°S is after Olierook et al. (2016). The Australian and Antarctica plates are put closer within permissible limits of geologic and paleomagnetic results (IND = India, AUS = Australia, and ANT = Antarctica). (b) The mean paleomagnetic pole data of the Rajmahal Traps plotted along the synthetic APWP for India based on the GAPWaP model by Torsvik et al. (2012). The black dot depicted with the value of A95 in a yellow circle between 110 and 120 Ma interval is the obtained VGP from this study [11.37°N, 297.58°E (62.42°W); A95 = 3.5°, k = 99.99]. We have depicted the other study results from RT and ST (of the RBS volcanic province) for comparison. The 'I' is a VGP from ST (16°N, 60°W; A95 = 7°; Athavale et al., 1963), 'II' is a VGP from ST (12°N, 59°W; A95 = 7°; Rao et al., 1993), 'III' is a VGP from RT (7°N, 63°W; A95 = 4°; Klootwijk, 1971), and 'IV' is a VGP from RT (3°N, 62°W; A95 = 4°; McDougall and McElhinny, 1970). The gray-colored circles indicate A95, around mean pole position. 'A' corresponds to the Deccan Super Pole (DSP, 36.9°N; 78.7°W) derived by Vandamme et al. (1991). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

The paleolatitude obtained from this study is 43.1°S, which is at least 6° north of the present-day location of Kerguelen hotspot (latitude = ~49.2°S), lying beneath the Kerguelen Plateau. It indicates a southward movement of the Kerguelen hotspot (mantle plume) by some 6° during the past ~ 117 Ma, which supports the findings of Mallik et al. (1999). Many studies are available (Torsvik et al., 1998, 2012; Steinberger and O'Connell, 2000; Antretter et al., 2002; Doubrovine et al., 2012) that dealt with the absolute motion or fixity of the Indian Ocean hotspots. The Marion hotspot (plume) exhibits southward drift of around 5-10° (Steinberger and O'Connell, 2000) similarly, the Reunion hotspot (plume) has drifted 5° northward (Vandamme and Courtillot, 1990). The Kerguelen hotspot which is the interest of this study has a 5-10° southward drift for last ~ 117 Ma (Su et al., 1994; Grand et al., 1997; Lithgow-Bertelloni and Richards, 1998; Gaina et al., 2003), and is consistent with our results. In summary, the interpretations from the present study are in good agreement with the literature and the predictions of numerical models of global mantle circulation (Steinberger and O'Connell, 1998; O'Neill et al., 2003; Steinberger et al., 2004).

The paleolatitude position and APW path from the present study are in a better agreement with the published literature and allowed us to infer the paleogeographic position of the Indian subcontinent and the migration of the Kerguelen hotspot (mantle plume) overtime. It is highly recommended that extensive flow-wise sample collection is necessary to light upon the magnetostratigraphic aspect and ascertain the polarity history of the Rajmahal Traps.

6. Conclusions

Paleomagnetic and rock magnetic investigations on RT resulted in a reliable dataset, and the following points are observed:

1. The main magnetic carrier mineral in these samples is a fine-grained, single domain (SD) magnetite and minerals ilmenite, pyrrhotite, and maghemite are in accessory form.
2. The mean ChRM direction give declination of 317.51° and inclination of −61.92° (α_{95} = 3.6°; k = 93.39).
3. The obtained virtual geomagnetic pole (VGP) position of the RT at ~ 117 Ma is 11.37°N, 297.58°E (62.42°W) with A95 = 3.5°, k = 99.99, dp = 4.33, and dm = 5.58, agrees well with synthetic APWP for India.
4. The calculated paleolatitude is at 43.1°S, suggest a mid-latitude southern hemispheric location of the Indian subcontinent at the time of formation of the Rajmahal Traps.
5. The present study results agree with anticipated global mantle circulations and suggest the migration of Kerguelen hotspot (mantle plume) southward by at least 6°.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors thank V.M. Tiwari (Director, CSIR-NGRI) for the encouragement and permission to publish this work (NGRI/Lib/2020/Pub-05). MRK especially thank A.P. Singh, Ravi Shankar, Ranjit Kumar Singh, Santosh Kumar Mandal, Lakshman Murmu, and all other field friends. The authors extend thanks to Manoj Pandit, Associate Editor, for helpful handling of the manuscript and anonymous reviewers for their critical and constructive comments that improved the manuscript. MRK extends thanks to CSIR, India, for the fellowship SRF (P-81-101).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jseaes.2020.104517.

References

Aitchison, J.C., Ali, J.R., Davis, A.M., 2007. When and where did India and Asia collide? J. Geophys. Res. 112, 1–19. https://doi.org/10.1029/2006JB004706.
Antretter, M., Steinberger, B., Heider, F., Soffel, H., 2002. Paleolatitudes of the Kerguelen hotspot: new paleomagnetic results and dynamic modeling. Earth Planet. Sci. Lett. 203, 635–650. https://doi.org/10.1016/S0012-821X(02)00841-5.
Athavale, R.N., Radhakrishnamurthy, C., Sahastrabudhe, P.W., 1963. Paleomagnetism of some Indian rocks. Geophys. J. R. Astron. Soc. 7, 304–313.
Bagchi, D., 1991. Magnetic properties and iron-titanium oxide mineralogy of some Rajmahal volcanic rocks, a. Ph.D. Thesis. University of Jadavpur, Calcutta, India.
Ball, V., 1877. Geology of the Rajmahal Hills. Mem. Geol. Surv. India 13, 155–248.
Baksi, A.K., Ray Barman, T., Paul, D.K., Farrar, E., 1987. Widespread Early Cretaceous flood basalt volcanism in eastern India: geochemical data from the Rajmahal-Bengal-Sylhet Traps. Chem. Geol. 63, 133–141.
Baksi, A.K., 1995. Petrogenesis and timing of volcanism in the Rajmahal flood basalt province, Northeastern India. Chem. Geol. 121, 73–90.
Mallik, S.B., Piper, J.D.A., Das, A.K., Bandyopadhyay, G., Sherwood, G.J., 1999. Paleomagnetic and rock magnetic studies in the Gondwana supergroup (Carboniferous–Cretaceous) northeast India. Geol. Soc. India Mem. 44, 87–116.
Bian, W., Yang, T., Ma, Y., Jin, J., Gao, F., Wang, S., Peng, W., Zhang, S., Wu, H., Li, H., Cao, L., Shi, Y., 2019. Paleomagnetic and geochronological results from the Zhela and Weimei Formations lava flows of the eastern Tethyan Himalaya: New insights into the break-up of eastern Gondwana. J. Geophys. Res.: Solid. Earth 124, 44–64. https://doi.org/10.1029/2018JB016403.
Clegg, J.A., Radhakrishnamurthy, C., Sahastrabudhe, P.W., 1958. The Remanent magnetism of the Rajmahal traps of north-eastern India. Nature 181, 830–831.

Coffin, M.F., 1992. Emplacement and subsidence of Indian Ocean plateaus and submarine ridges. In: In: Duncan, R.A., Rea, D.K., Kidd, R.B., von Rad, U., Weissel, J.K. (Eds.), Synthesis of Results from Scientific Drilling 70. Am. Geophys. Union Monogr, pp. 115–125.

Coffin, M.F., Pringle, M.S., Duncan, R.A., Gladczenko, T.P., Storey, M., Muller, R.D., Gahagan, L.A., 2002. Kerguelen Hotspot Magma Output since 130 Ma. J. Petrology 43 (7), 1121–1137. https://doi.org/10.1093/petrology/43.7.1121.

Courtillot, V., 1999. Evolutionary Catastrophes: The Science of Mass Extinctions. Press p, Camb. Uni, pp. 95.

Das, A.K., Piper, J.A., Bandyopadhyay, G., Mallik, S.B., 1996. Polarity inversion in the Rajmahal lavas, north-east India: trap emplacement near commencement of the Cretaceous Normal Superchron. Geophys. J. Int. 124, 427–432.

Davies, H.L., Sun, S.S., Frey, F.A., Gautier, I., McCulloch, M.T., Price, R.C., Bassias, Y., Klootwijk, C.T., Leclaire, L., 1989. Basalt basement from the Kerguelen Plateau and the trail of a DUPAL plume. Contrib. Mineral. Petrol. 103, 457–469.

Day, R., Fuller, M.D., Schmidt, V.A., 1977. Hysteresis properties of titanomagnetites: grain size and composition dependence. Phys. Earth Planet. Inter. 13, 260–266.

Deenen, M.H.L., Langereis, C.G., Hinsbergen van, D.J.J., Biggin, A.J., 2011. Geomagnetic secular variation and the statistics of palaeomagnetic directions. Geophys. J. Int. 186, 509–520. https://doi.org/10.1111/j.1365-246X.2011. 05050.x.

Doubrovine, P.V., Steinberger, B., Torsvik, T.H., 2012. Absolute plate motions in a reference frame defined by moving hot spots in the Pacific, Atlantic, and Indian oceans. J. Geophys. Res. 117, B09101. https://doi.org/10.1029/02011JB009072.

Duncan, R.A., 1981. Hotspots in the southern oceans- an absolute frame of reference for motion of Gondwana continents. Tectonophysics 74, 29–42.

Dunlop, D.J., Ozdemir, O., 1997. Rock Magnetism, Fundamentals, and Frontiers. Cambridge University Press, Cambridge 1–595.

Dunlop, D.J., 2002. Theory and application of the day plot (Mrs/Ms versus Hcr/Hc) 2. Application to data for rocks, sediments, and soils. J. Geophys. Res. 107, B3, 2057, EPM5-1- EPM5-15. https://doi.org/10.1029/2001JB000487.

Fisher, R.A., 1953. Dispersion on a sphere. Proc. Royal. Soc. London A 217, 295–305.

Frey, F.A., Coffin, M.F., Wallace, P.J., Weis, D., Zhao, X., Wise Jr, S.W., Wahnert, V., Teagle, D.A.H., Saccocia, P.J., Reusch, D.N., Pringle, M.S., Nicolaysen, K.E., Neal, C.R., Muller, R.D., Moore, C.L., Mahoney, J.J., Keszthelyi, L., Inokuchi, H., Duncan, R.A., Delius, H., Damuth, J.E., Damasceno, D., Coxall, H.K., Borre, M.K., Boehm, F., Barling, J., Arndt, N.T., Antretter, M., 2000. Origin and evolution of a submarine large igneous province: the Kerguelen Plateau and Broken Ridge, southern Indian Ocean. Earth Planet. Sci. Lett. 176, 73–89.

Gaina, C., Muller, R.D., Brown, B., Ishihara, T., 2003. Microcontinent formation around Australia. In: R. Hillis, R.D. Muller (Eds.), The Evolution and Dynamics of the Australian Plate, Geol. Soc. Aust. Spec. Publ. 22 and Geol. Soc. Am. Spec. Paper 372, 399-410.

Gaina, C., Muller, R.D., Brown, B., Ishihara, T., 2007. Breakup and early seafloor spreading between India and Antarctica. Geophys. J. Int. 170, 151–169.

Ghatak, A., Basu, A.R., 2013. Isotopic and trace element geochemistry of alkalic–mafic-ultramafic-carbonatitic complexes and flood basalts in NE India: Origin in a heterogeneous Kerguelen Plume. Geochimica et Cosmochimica Acta 115, 46–72.

Ghosh, S., 1979. The magnetisation of the volcanic flows from Northeastern Rajmahal Hills, Bihar, India, a. Ph.D. Thesis. University of Jadavpur, Calcutta, India.

Grand, S.P., van der Hilst, R.D., Widiyantoro, S., 1997. Global seismic tomography: A snapshot of convection in the Earth. GSA Today 7, 1–7.

Heine, C., Muller, R.D., Gaina, C., 2004. Reconstructing the Lost Eastern Tethys Ocean Basin: Convergence history of the SE Asian margin and marine gateways. In: Continent- Ocean Interactions in Southeast Asia, eds. P. Clift, D. Hayes, W. Kuhnt, and P. Wang. Am. Geophys. Union Monogr. 149, 37-54.

Hobson, G.V., 1929. General Report. Geol. Surv. India Records of the Geol. Survey India, LXII 1, 145–146.

Kapawar, M.R., Venkateshwarlu, M., 2019. Rock Magnetic and paleomagnetic investigations of Sylhet traps, Shillong Plateau. NE India. J. Geodynamics 127, 31–41. https://doi.org/10.1016/j.jog.2019.05.003.

Kent, R., 1991. Lithospheric uplift in eastern Gondwana: evidence for long-lived mantle plume system? Geology 19, 19–23.

Kent, R.W., Pringle, M.S., Muller, R.D., Saunders, A.D., Ghose, N.C., 2002. $^{40}Ar/^{39}Ar$ geochronology of the Rajmahal basalts, India, and their relationship to the Kerguelen Plateau. J. Petrology 43, 1141–1154.

Kirschvink, J.L., 1980. The least-squares line and plane and the analysis of paleomagnetic data. Geophys. J. Roy. Astron. Soc. 62, 699–718. https://doi.org/10.1111/j.1365246X. 1980.tb02601.x.

Klootwijk, C.T., 1971. Palaeomagnetism of the Upper Gondwana Rajmahal Traps, Northeast India. Tectonophysics 12, 449–467.

Klootwijk, C.T., Gee, J.S., Peirce, J.W., Smith, G.M., 1992. An early India-Asia contact: Paleomagnetic constraints from Ninetyeast Ridge, ODP Leg 121. Geology 20, 395–398.

Lithgow-Bertelloni, C., Richards, M.A., 1998. The dynamics of Cenozoic and Mesozoic plate motions. Rev. Geophys. 36, 27–78. https://doi.org/10.1029/97RG02282.

Luyendyk, B.P., Rennick, W., 1977. Tectonic history of the aseismic ridges in the eastern Indian ocean. Geol. Soc. Amer. Bull. 88, 1347–1356.

Mahoney, J.J., McDougall, J.D., Lugmair, G.W., Gopalan, K., 1983. Kerguelen hotspot source for Rajmahal Traps and Ninetyeast Ridge? Nature 303, 385–389.

McDougall, I., McElhinny, M.W., 1970. The Rajmahal Traps of India- K-Ar ages and Palaeomagnetism. Earth Planet. Sci. Lett. 9, 371–378.

Meert, J.G., Tamrat, E., 2006. Paleomagnetic evidence for a stationary Marion hotspot: Additional paleomagnetic data from Madagascar. Gondwana Res. 10, 340–348. https://doi.org/10.1016/j.gr.2006.04.008.

Mihut, D., Muller, R.D., 1998. Volcanic margin formation and Mesozoic rift propagators in the Cuvier Abyssal Plain off Western Australia. J. Geophys. Res. 103, 27135–27149. https://doi.org/10.1029/97JB02672.

Morgan, W.J., 1981. Hotspot tracks and the opening of the Atlantic and Indian ocean. In: Emiliani, C. (Ed.), The Sea, 7- The Oceanic Lithosphere. Wiley, New York, pp. 443–487.

Olierook, H.K.H., Jourdan, F., Merle, R.E., Timms, N.E., Kusznir, N., Muhling, J., 2016. Bunbury Basalt: Gondwana break-up products or earliest vestiges of the Kerguelen mantle plume? Earth Planet. Sci. Lett. 440, 20–32. https://doi.org/10.1016/j.epsl.2016.02.008.

O'Neill, C., Muller, D., Steinberger, B., 2003. Geodynamic implications of moving Indian ocean hotspots. Earth Planet. Sci. Lett. 215, 151–168. https://doi.org/10.1016/S0012-821X(03)00368-6.

Pascoe, E., 1959. A manual of Geology of India and Burma, Gov 3, 485–1343.

Qin, S.X., Li, Y.X., Li, X.H., Xu, B., Luo, H., 2019. Paleomagnetic results of Cretaceous cherts from Zhongba, southern Tibet: New constraints on the India-Asia collision. J. Asian Earth Sci. 173, 42–53. https://doi.org/10.1016/j.jseaes.2019.01.012.

Radhakrishna, T., Joseph, M., 2012. Geochemistry and paleomagnetism of late Cretaceous mafic dikes in Kerala, southwest coast of India in relation to large igneous provinces and mantle plumes in the Indian Ocean region. Geol. Soc. Am. Bull. 124 (1/2), 240–255. https://doi.org/10.1130/b30288.1.

Radhakrishnamurthy, C., 1963. Remanent magnetisation of the igneous rocks in the Gondwana Formation of India, a. Ph.D. Thesis. Andhra University, India.

Radhakrishnamurthy, C., 1970. In: Laboratory studies for ascertaining the suitability of rocks for palaeomagnetism. Academic Press, London, pp. 235–241.

Radhakrishnamurthy, C., Likhite, S.D., Sahastrabudhe, P.W., 1977. Nature of magnetic grains and their effect on the remanent magnetisation of basalts. Phys. Earth Planet. Inter. 13, 289–300.

Radhakrishnamurthy, C., Likhite, S.D., Sahastrabudhe, P.W., 1991. Domain states of magnetic grains in basalts and palaeointensity techniques. J. Geomag. Geoelectr. 43, 351–355.

Rao, G.V.S.P., Rao, M.V.S., Rao, J.M., 1993. Geochemical and palaeomagnetic signatures in Sylhet Trap flood basalt volcanism: preliminary results. Geol. Mag. 130, 163–184.

Rao, G.V.S.P., Rao, J.M., Rao, M.V.S., 1996. Palaeomagnetic and geochemical characteristics of the Rajmahal Traps, eastern India. J. Southeast Asian Earth Sci. 13 (2), 113–122.

Rao, G.V.S.P., Rao, J.M., 1996. Paleomagnetism of the Rajmahal Traps of India: implication to the reversal in the Cretaceous Normal Superchron. J. Geomag. Geoelectr. 48, 993–1000.

Roy Chowdhury, P.E., 1965. Bihar District Gazetteers Santhal Paragana 7–9, and 36..

Royer, J.Y., Coffin, M.F., 1992. Jurassic to Eocene plate tectonic reconstructions in the Kerguelen Plateau region. Proc. Ocean Drill. Prog. Sci. Results 120, 917–928.

Sherwood, G.J., 1992. Some rock magnetic properties of mid-Cretaceous basalts from Israel and India (Rajmahal traps), and their bearing on palaeointensity experiments. Phys. Earth Planet. Inter. 70, 237–242.

Sherwood, G.J., Baer, G., Mallik, S.B., 1993. The strength of the geomagnetic field during the Cretaceous Quiet Zone: palaeointensity results from Israeli and Indian lavas. J. Geomag. Geoelectr. 45, 339–360.

Sherwood, G.J., Mallik, S.B., 1996. A palaeomagnetic and rock magnetic study of the northern Rajmahal Volcanics, Bihar. India. J. Southeast Asian Earth Sci. 13 (2), 123–131.

Steinberger, B., O'Connell, R.J., 1998. Advection of plumes in mantle flow: Implications for hotspot motion, mantle viscosity and plume distribution. Geophys. J. Int. 132, 412–434. https://doi.org/10.1046/j.1365-246x.1998.00447.x.

Steinberger, B., O'Connell, R.J., 2000. Effects of mantle flow on hotspot motion. In: Richards, M.A., Gordon, R.G., van der Hilst, R.D. (Eds.), The History and Dynamics of Global Plate Motions. AGU, Washington, DC, pp. 377–398.

Steinberger, B., Sutcherland, R., O'Connell, R.J., 2004. Prediction of Emperor-Hawaii seamount locations from a revised model of global plate motion and mantle flow. Nature 430, 167–173. https://doi.org/10.1038/nature02660.

Su, W.J., Woodward, R.L., Dziewonski, A.M., 1994. Degree 12 model of shear velocity heterogeneity in the mantle. J. Geophys. Res. 99, 6945–6980.

Tiwari, S., Jassal, G.S., 2003. Origin and evolution of the Garo-Rajmahal Gap. J. Geol. Soc. India 57, 389–403.

Torsvik, T.H., Tucker, R.D., Ashwal, L.D., Eide, E.A., Rakotosolofo, N.A., de Wit, M.J., 1998. Late Cretaceous magmatism in Madagascar: palaeomagnetic evidence for a stationary Marion hotspot. Earth Planet. Sci. Lett. 164, 221–232.

Torsvik, T.H., Muller, R.D., Van der Voo, R., Steinberger, B., Gaina, C., 2008. Global plate motion frames: toward a unified model. Rev. Geophys. 46 10.1029/2007RG000227.RG3004.

Torsvik, T.H., Van der Voo, R., Preeden, U., Mac Niocaill, C., Steinberger, B., Doubrovine, P.V., van Hinsbergen, D.J.J., Domeier, M., Gaina, C., Tohver, E., Meert, J.G., McCausland, P.J.A., Cocks, L.R.M., 2012. Phanerozoic polar wander, palaeogeography, and dynamics. Earth Sci. Rev. 114, 325–368. https://doi.org/10.1016/j.earscirev.2012.06.007.

Vandamme, D., Courtillot, V., 1990. Paleomagnetism of Leg 115 basement rocks and latitudinal evolution of the Reunion hotspot. In: R. A. Duncan, J. Backman, R. B. Dunbar, L.C. Peterson (Eds.), Proc. ODP Sci. Results 115, 111-118.

Vandamme, D., Courtillot, V., Besse, J., Montigny, R., 1991. Paleomagnetism and age determinations of the Deccan traps (India): results of a Nagpur-Bombay traverse and review of earlier work. Rev. Geophys. 29, 159–190. https://doi.org/10.1029/91RG00218.

Van der Voo, R., 1990. The reliability of paleomagnetic data. Tectonophysics 184, 1–9.

Zijderveld, J.D.A., 1967. AC demagnetization of rocks: analysis of results. In: Collinson, D.W., Creer, K.M., Runcorn, S.K. (Eds.), Methods in Paleomagnetism. Elsevier, Amsterdam, pp. 254–286.